Quantum Mechanics I HW 6

Pick 4 out of the 5 questions to answer

Question 1)

a) Starting from the definition of the time-dependent operator in the Heisenberg picture, derive the Heisenberg equation of motion. You can assume $\hat{H}$ is time-independent.

b) By taking the expectation value of the Heisenberg equation of motion, write the corresponding equation for the expectation value of the operator. Does it depend on what picture (Heisenberg or Schrödinger) you are using?

c) Let $\hat{W}(\lambda) = e^{-\frac{i}{\hbar}\lambda\hat{G}}$, with $\hat{G}$ Hermitian and possibly explicitly time-dependent. We call $\hat{W}$ a symmetry if, for every λ, it maps solutions of the Schrödinger equation to the same or other solutions of the Schrödinger equation. That is, if $|\psi(t)\rangle$ is a solution, then $\hat{W}(\lambda)\,|\psi(t)\rangle$ should also be a solution for $\hat{W}(\lambda)$ to be called a symmetry. **Without using the Heisenberg picture**, show that $\hat{W}$ is a symmetry if $i\hbar\,\partial_t\hat{W} = [\hat{H}, \hat{W}]$. Recall that a solution respects:

$$i\hbar\frac{\partial}{\partial t}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle\,.$$

Is this consistent with the Heisenberg equation of motion?

Question 2) Neutrinos oscillate between three flavors ν_e, ν_μ, and ν_τ. The flavor states are given by

$$|\nu_\alpha\rangle = \sum_{i=1}^{3} U^*_{\alpha i}\,|\nu_i\rangle\,, \quad \alpha = e, \mu, \tau, \quad i = 1, 2, 3 \tag{1}$$

where U is the unitary mixing matrix that relates the flavor eigenstates $|\nu_\alpha\rangle$ to the mass/energy eigenstates $|\nu_i\rangle$. In the energy basis, $\hat{H}\,|\nu_i\rangle = E_i\,|\nu_i\rangle$. Assume a non-degenerate spectrum of energies, $E_i \neq E_j$ for $i \neq j$.

a) Write the probability for a neutrino to oscillate from an initial flavor state $|\nu_\alpha\rangle$ to a final flavor state $|\nu_\beta\rangle$ at a later time t. Express your answer only in terms of the mixing matrix elements $U_{\alpha i}$ and complex exponentials of differences of energies, $\Delta E_{ij} = E_i - E_j$.

b) What is the oscillation probability when the matrix $U_{\alpha i}$ is diagonal?

c) The IceCube experiment detects neutrinos from astrophysical sources, megaparsecs to maybe gigaparsecs away. Using your answer from part (a), do you expect IceCube to observe neutrinos in the same flavor composition as they were produced?

d) Assume that the unitary mixing matrix is fully democratic

$$|U_{\alpha i}| = \frac{1}{\sqrt{3}} \quad \text{for all } \alpha \text{ and } i. \tag{2}$$

Averaging over time at $t \to \infty$, what is the expected flavor composition of neutrinos detected by IceCube as a ratio of $\nu_e : \nu_\mu : \nu_\tau$? Does it depend on the initial flavor composition?

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Question 3) Consider a free particle in a one-dimensional space:

$$\hat{H} = \frac{\hat{p}^2}{2m}. \tag{3}$$

The generator of Galilean boosts is given by

$$\hat{K} = m\hat{x} - t\hat{p}, \tag{4}$$

with t time, $\hat{x}$ the position operator, and $\hat{p}$ the momentum operator.

a) Start by considering $\hat{K}$ as a generator of a unitary transformation (Galilean boost). Show that for small boosts by a small velocity v, the momentum operator transforms as

$$\left(\mathbb{1} - \frac{i}{\hbar}v\hat{K}\right)\hat{p}\left(\mathbb{1} + \frac{i}{\hbar}v\hat{K}\right) = \hat{p} + vm + \mathcal{O}(v^2) \tag{5}$$

and the position operator transforms as

$$\left(\mathbb{1} - \frac{i}{\hbar}v\hat{K}\right)\hat{x}\left(\mathbb{1} + \frac{i}{\hbar}v\hat{K}\right) = \hat{x} + vt + \mathcal{O}(v^2) \tag{6}$$

b) Consider $\hat{x}$, $\hat{p}$, $\hat{H}$, and $\hat{K}$. Which of these Heisenberg picture operators are constant in time for the free particle?

c) Using your result above and the initial conditions for $\hat{x}(0) = \hat{x}_0$ and $\hat{p}(0) = \hat{p}_0$, where

$$\hat{K}_0 = m\hat{x}_0 \tag{7}$$

determine $\hat{x}(t)$ in terms of $\hat{p}_0$, $\hat{x}_0$, m, and t.

Question 4)

a) **Briefly** describe the Stern-Gerlach experiment and how it selects spin states.

b) Starting from an unpolarized collection of spins traveling in the y direction, we place a Stern-Gerlach filter along the z-axis (SG$z(+)$). What spin eigenstate is selected by this filter?

c) We now place a Stern-Gerlach filter along the x-axis (SG$x(+)$). How does the intensity of the beam change compared to the beam that **enters** the SG$z(+)$ filter?

d) Now place a third Stern-Gerlach filter along the z-axis:

$$\text{SG}z(+) \to \text{SG}x(+) \to \text{SG}z(-) \tag{8}$$

What is the beam intensity after passing through the third filter compared to the initial unpolarized beam? What happens if we remove the second filter (SG$x(+)$)?

Question 5) Consider the two-level system of a spin-1/2 particle in a magnetic field pointing along the z-axis:

$$\hat{H} = \omega\hat{S}_z = \frac{\omega}{2}\hbar\sigma_z, \quad \omega = |e|B/m_e,$$

where B is the magnitude of the magnetic field.

a) For the above Hamiltonian, working in the Heisenberg picture, write the equation of motion for the spin operators $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$. Recall the Heisenberg equation of motion for an operator $\hat{O}$:

$$\frac{d\hat{O}}{dt} = \frac{1}{i\hbar}[\hat{O}, \hat{H}] + \frac{\partial \hat{O}}{\partial t}.$$

b) Differentiate the spin operators $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$ again and solve the second-order differential equations to find $\hat{S}_x(t)$, $\hat{S}_y(t)$, and $\hat{S}_z(t)$ as a function of $\hat{S}_x(0)$, $\hat{S}_y(0)$, and $\hat{S}_z(0)$. You will need the first and second derivatives of the spin operators to simplify things.

c) Discuss your result: what is the Bloch vector of this system and how does it evolve in time for this spin-1/2 particle in the magnetic field?

Formulae

- Any two-state system state can be as

$$|\psi\rangle = \cos\frac{\theta}{2}|+\rangle + e^{i\phi}\sin\frac{\theta}{2}|-\rangle$$

where $S_z|\pm\rangle = \pm\frac{\hbar}{2}|\pm\rangle$.

- $[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$, with ϵ_{ijk} the Levi-Civita symbol ($i, j, k = x, y, z$, $\epsilon_{xyz} = +1$).
- $S_i = \frac{\hbar}{2}\sigma_i$, with σ_i the Pauli matrices. $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
- Time evolution operator:

$$\hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}t}.$$

- Heisenberg picture time evolution:

$\hat{O}_H(t) = e^{\frac{i}{\hbar}\hat{H}t}\hat{O}_S e^{-\frac{i}{\hbar}\hat{H}t}$ where $\hat{O}_S$ is the Schrödinger picture operator.

- Schrodinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle, \quad i\hbar\frac{\partial}{\partial t}\hat{U}(t) = \hat{H}\hat{U}(t)$$

- Heisenberg equation of motion:

$$\frac{d\hat{O}_H(t)}{dt} = \frac{1}{i\hbar}[\hat{O}_H(t), \hat{H}] + \left(\frac{\partial \hat{O}(t)}{\partial t}\right)_H$$